Artificial intelligence (AI)

&

machine learning (ML)

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- **INTELLIGENCE**

- Who is intelligent?
- All living organisms are intelligent.
- They interact with their environment and survive.
- Examples from our own world

➢Crossing a road

➢Discovering alternate paths

• ARTIFICIAL INTELLIGENCE

• Living beings are intelligent; but are man made non living beings also intelligent???

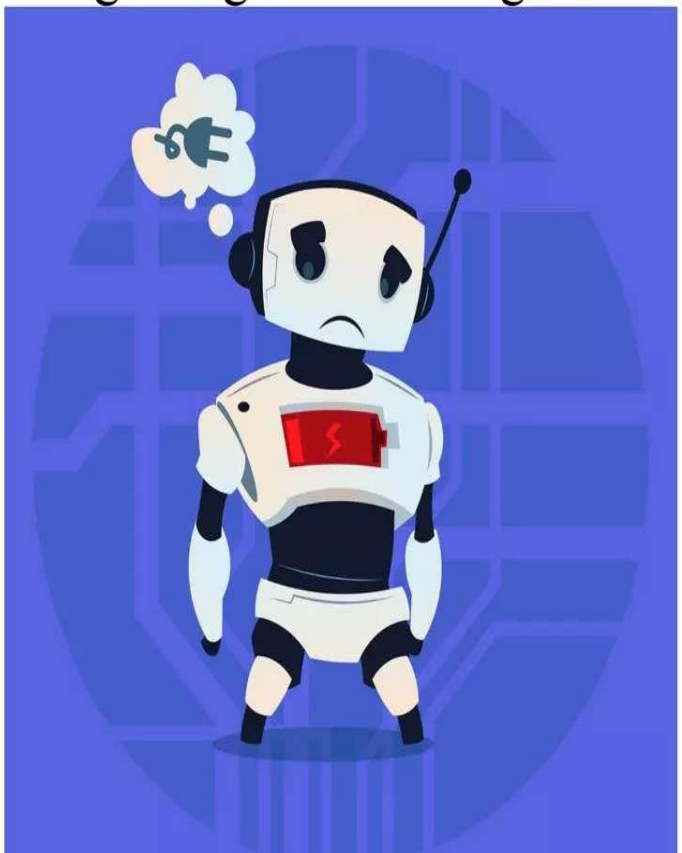
• Can a machine

➢make discoveries?

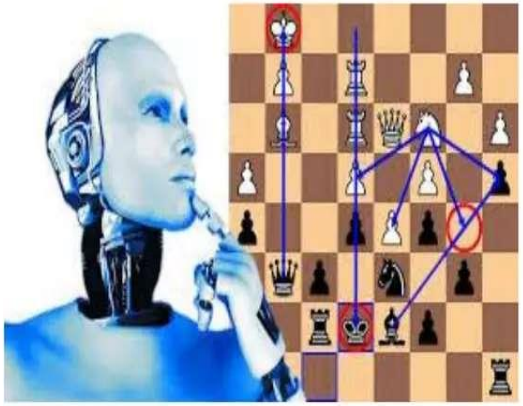
➢pass a ruling order in a court?

➢compose a symphony?

➢go for a PLAN B?

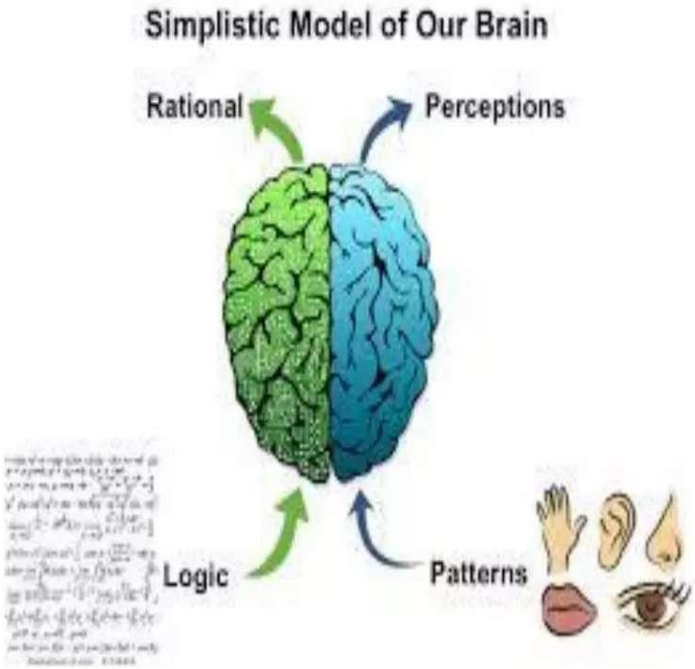


- **ARTIFICIAL INTELLIGENCE**



- Traditional computers are powerful but not intelligent
- They can compile MBs and GBs of code but may get stuck at a minor logical error
- Artificial intelligence is a field of computer science which aims to make computer systems that can mimic human intelligence.
- Just as we humans act when we don't have exact information about a

- **ARTIFICIAL INTELLIGENCE**



- Why make machines INTELLIGENT?
- To reduce our effort and help the society advance

➢share our load

➢make use of massive number crunching power of CPUs

➢perceive things and try to realize them

➢perform in our absence/ without our guidance

- **ARTIFICIAL INTELLIGENCE SUBSETS**

- MACHINE LEARNING
- ARTIFICIAL NEURAL NETWORKS
- DEEP LEARNING
- COMPUER VISION
- NATURAL LANGUAGE PROCESSING
- SPEECH RECOGNITION

- **MACHINE LEARNING**

- It is a branch of Artificial Intelligence that gives computers the capability to learn without being explicitly programmed.
- Focus is on imparting “learning” to machines
- Learning over time and iterations (similar to human experience)
- No longer dependent on rule based programming
- Real world data and observations are fed to the system

- **MACHINE LEARNING**

- ML algorithms can be broadly categorized into

➢SUPERVISED

➢UNSUPERVISED

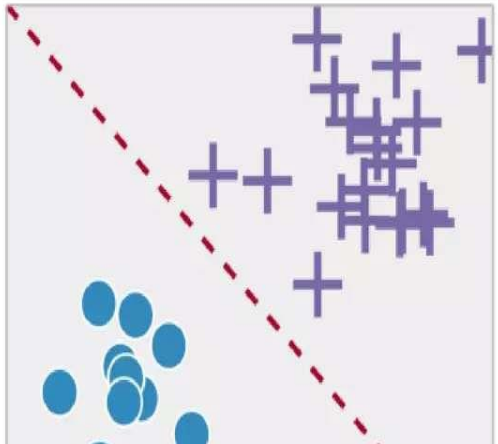
➢REINFORCED

- **MACHINE LEARNING**

- **SUPERVISED LEARNING**
- Uses ground truth and labeled data
- Requires prior knowledge
- Approximates the relationship between input and output
- Mainly divided into CLASSIFICATION and REGRESSION
- Naïve Bayes, Random Forest, Support Vector Machine, Neural Networks

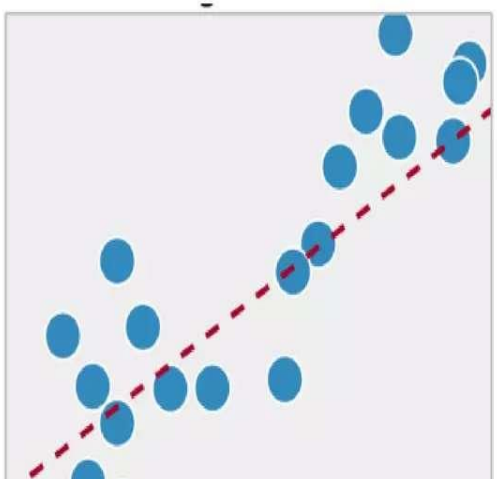
- **MACHINE LEARNING (SUPERVISED)**

- **CLASSIFICATION**
- approximating a mapping function (f) from input variables (X) to discrete output variables (y)
- Predicting a label
- Spam/ non spam
- Positive/ negative



- **MACHINE LEARNING (SUPERVISED)**

- **REGRESSION**
- Approximating a mapping function (f) from input variables (X) to a continuous output variable (y)
- Predicting a quantity
- Predict salary from age/experience data
- Sales forecast



- **MACHINE LEARNING**

- **UNSUPERVISED LEARNING**
- No historical labels
- Learn the inherent structure of data
- Discover the trends in data
- Mainly divided into CLUSTERING and ASSOCIATION

- **MACHINE LEARNING (UNSUPERVISED)**

- **CLUSTERING**
- Dividing the population into groups
- Same group members resemble each other compared to other groups
- Connectivity/ centroid/ distribution/density models
- K Means, Hierarchical, KNN, PCA

- **MACHINE LEARNING (UNSUPERVISED)**

- **ASSOCIATION**

- Rule based learning model

- Discover rules that describe large portions of your data

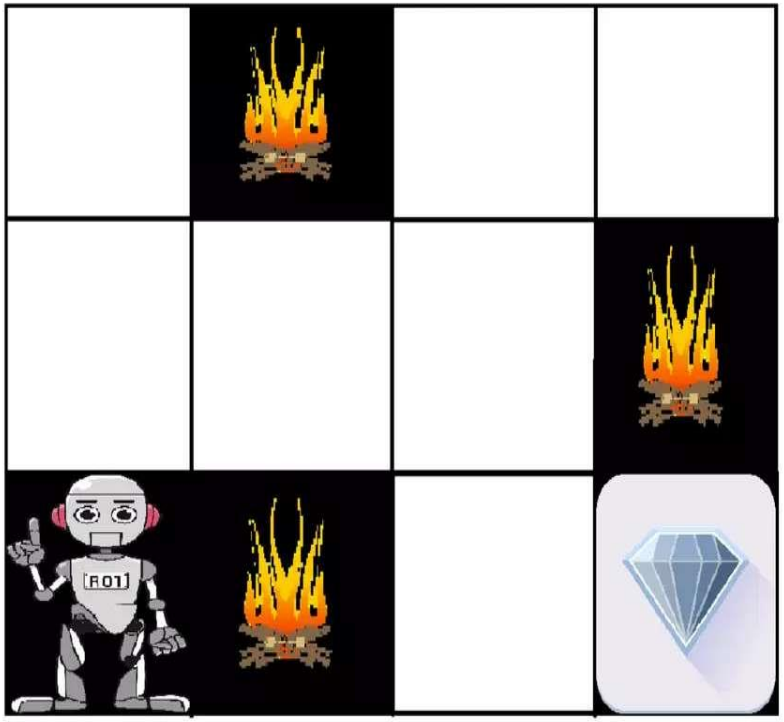
- Product placement in malls

- Eg people that buy X also tend to buy Y

ID	Items
1	{Bread, Milk}
2	{Bread, Diapers, Beer, Eggs}
3	{Milk, Diapers, Beer, Cola}
4	{Bread, Milk, Diapers, Beer}
5	{Bread, Milk, Diapers, Cola}

market basket transactions

- **MACHINE LEARNING**



- **REINFORCEMENT**
- Maximize reward in a given situation
- Find the best possible behavior/ path
- Input: initial state of the model
- Output: many possible solutions to a given problem
- Training: reward or punishment

- **MACHINE LEARNING TOOLS**

- PYTHON
- MATLAB
- R
- KNIME
- WEKA
- PYTORCH
- GEPHI
- NODEXL

• APPLICATIONS OF MACHINE LEARNING

- NATURAL LANGUAGE PROCESSING
- SENTIMENT ANALYSIS
- HANDWRITING RECOGNITION
- SPEECH/ FACIAL RECOGNITION
- CUSTOMER PROFILING (banks and financial institutions)
- RECOMMENDATION SYSTEM (movie and e commerce)
- CUSTOMER CHURN PREDICTION (telecommunication)